

ETHOS x SAPTANG LABS

# Product Development Challenge

There is a **critical need** to develop a **privacy-aware, explainable monitoring system** that can:



**Blueprint** of the system – explaining **what** the major pieces are, **how** they fit together, and **why** they were designed that way.

MODULE 4

STORAGE

- **Firebase** - Storage
- **MongoDB** - Database

MODULE 3

ML

- **Face Detection:** MTCNN
- **Face Embedding:** keras-facenet
- **Similarity:** scipy (Cosine)

MODULE 2

BACKEND

- **Framework:** Flask 2.3+
- **WebSockets:** flask-socketio
- **Data:** pandas, numpy

MODULE 1

FRONTEND

- **Framework:** React.js 18+
- **Routing:** react-router-dom
- **Styling:** Tailwind CSS
- **Visualizations:** heatmap.js, chart.js
- **Real-time:** socket.io-client
- **Fuse.js**- incorporating fuzzy search logic

MODULE 5

AI

- **Gemini AI :**
- Natural language processing for timeline summaries

**Kafka pipeline** used for Real time Data Interpretation and Risk score generation

**NLU** for Query Answering

**XGBOOST Model** used for training on the Data using 6 Features

**Flask**, web framework to use the Python Code including the ML part

**APIs** for Integration GeminiAI and various other features

## TECH STACK



Flask/Express 

Firebase Storage 

**XGBoost**

React 

Gemini  
Summary  
Generation

Python  
 

Let the **numbers** speak about  
the implementation

Features

6

Reduced Complexity

F1 Score

0.95

Weighted Avg F1-Score

Model Accuracy

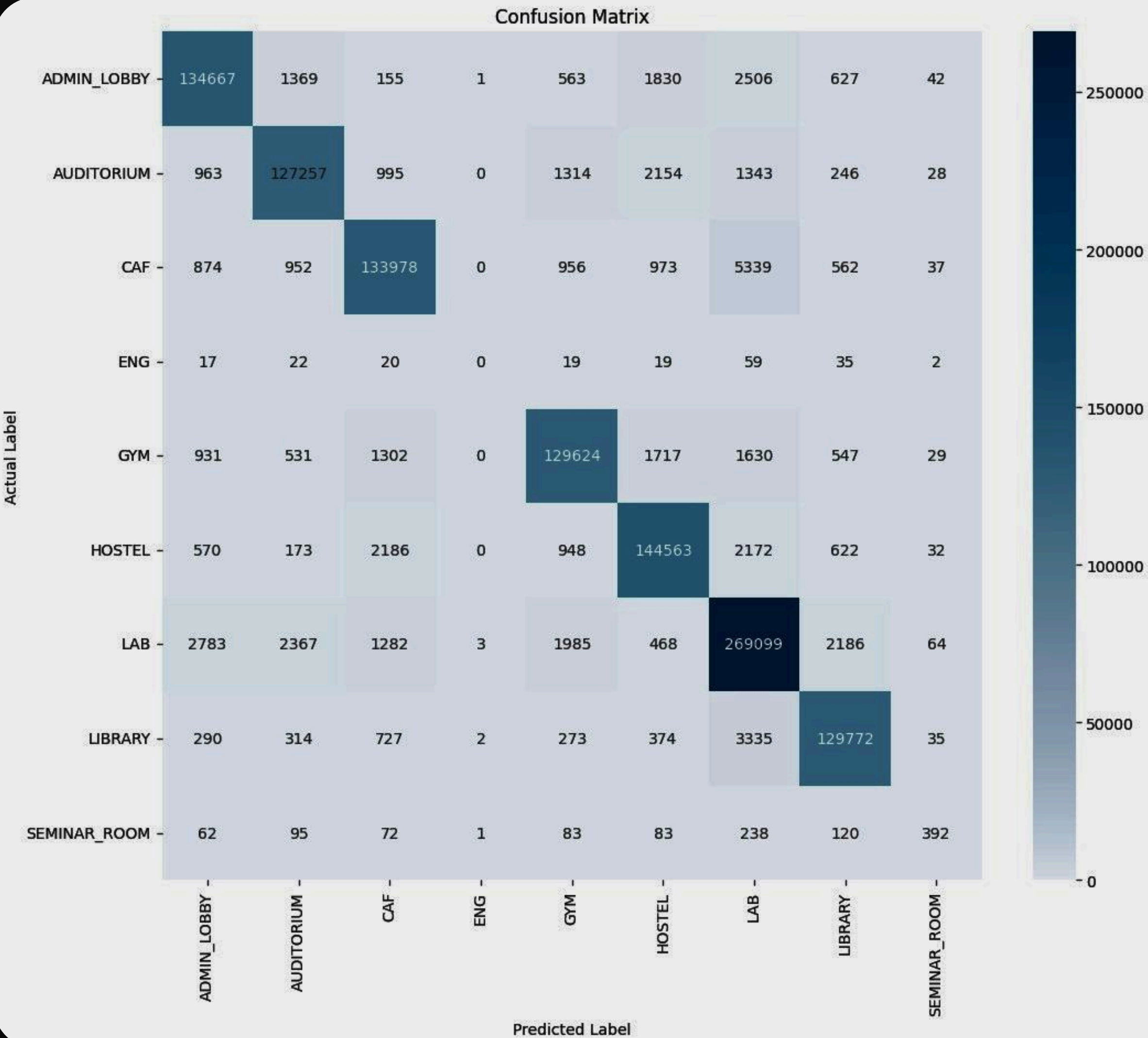
95.14%

XGBoost

Model Accuracy: 0.9514

Classification Report:

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| ADMIN_LOBBY  | 0.95      | 0.95   | 0.95     | 141760  |
| AUDITORIUM   | 0.96      | 0.95   | 0.95     | 134300  |
| CAF          | 0.95      | 0.93   | 0.94     | 143671  |
| ENG          | 0.00      | 0.00   | 0.00     | 193     |
| GYM          | 0.95      | 0.95   | 0.95     | 136311  |
| HOSTEL       | 0.95      | 0.96   | 0.95     | 151266  |
| LAB          | 0.94      | 0.96   | 0.95     | 280237  |
| LIBRARY      | 0.96      | 0.96   | 0.96     | 135122  |
| SEMINAR_ROOM | 0.59      | 0.34   | 0.43     | 1146    |
| accuracy     |           |        | 0.95     | 1124006 |
| macro avg    | 0.81      | 0.78   | 0.79     | 1124006 |
| weighted avg | 0.95      | 0.95   | 0.95     | 1124006 |



# Multi Modal Fusion Approach for using all the available data

MARKED USERS

Dashboard

Campus Activity

Data Upload Center

Upload data files to import them into the system.

Campus Card Swipes

Choose FileNo file chosen

Upload

Room Bookings

Choose FileNo file chosen

Upload

CCTV Frames

Choose FileNo file chosen

Upload

TrustUsBro

Dashboard

Campus Activity

Search

Alerts

Upload

Logout

Upload

Search

Alerts

Activity

Marked Users

Neha Mehta

Student

Last Seen:

ROOM A2

9/20/2025,  
9:38 AM

Expected Location:

LAB

Based on  
predictive model

Neha Gupta

Staff

Last Seen:

AUDITORIUM

9/21/2025,  
10:31 AM

Expected Location:

ADMIN LOBBY

Based on  
predictive model

Aarav Gupta

Student

Last Seen:

Library

9/21/2025,  
2:48 AM

Expected Location:

ADMIN LOBBY

Based on  
predictive model

Sana Desai

Student

Last Seen:

AP AP ADMIN 5

9/25/2025,  
9:05 AM

Expected Location:

ADMIN LOBBY

Based on  
predictive model

Divya Iyer

Student

Last Seen:

HOSTEL

10/25/2025,  
12:25 PM

Expected Location:

LIBRARY

Based on  
predictive model

Neha Kumar

Student

Last Seen:

AP LIBRARY

10/25/2025,  
11:40 AM

Expected Location:

ADMIN LOBBY

Based on  
predictive model

# User Dashboard with full details including the Timeline Generation Feature

user25@campus.edu

DH1b956db41369

Card Swipes (2)

Bookings (1)

CCTV Captures (0)

Wifi Logs (2)

Library Issues (0)

Notes (2)

| LOCATION    | TIMESTAMP              |
|-------------|------------------------|
| AUDITORIUM  | 9/21/2025, 10:31:58 AM |
| ADMIN LOBBY | 9/16/2025, 2:38:36 AM  |

Historical Activity Timeline

Location Timeline

Monthly

Weekly

1

6

11

16

21

26

Showing activity for September 2025

TrustUsBro

DashboardCampus ActivitySearchAlertsUploadLogout

Users / Neha Gupta

Neha Gupta

Staff

Unmark User

Personal Details

Entity ID

E100025

Department

MECH

Email

user25@campus.edu

Device Hash

DH1b956db41369

Card Swipes (2)

Bookings (1)

CCTV Captures (0)

Wifi Logs (2)

Library Issues (0)

Notes (2)

| LOCATION | TIMESTAMP |
|----------|-----------|
|----------|-----------|

MARK USER

MONTHLY / WEEKLY  
TIMELINE

ETHOS 25

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Fuzzy logic for *handling typos* in entity search

*NLU – Natural Language Understanding* for Query search

Dashboard

Can...

AskMeBro

How can I assist you today?

find user neha

Found 677 matching users:

Neha Mehta

ID: E100000 | Type: student

Neha Kumar

ID: E100001 | Type: student

Neha Singh

ID: E100002 | Type: student

Neha Rao

ID: E100005 | Type: student

Neha Malhotra

ID: E100010 | Type: student

Neha Gupta

ID: E100025 | Type: staff

e.g., find user Neha

Search

Search for users or data

Alerts

View active alerts

Expected Location: LAB

Based on predictive model

Neha Gupta

Staff

Last Seen: ADMIN L

10/25/2025, 10:24 PM

Expected Location: HOSTEL

Based on predictive model

Sana Desai

Student

Last Seen: AP AP AD

9/25/2025, 9:05 AM

Expected Location: AUDITORIUM

Based on predictive model

Neha Kumar

Student

Last Seen: ADMIN I

10/25/2025, 10:24 PM

Search Entities

Find students, faculty, or staff by a specific criterion.

Search By

Name

Search Value

neha

Search

Search Results

| Name          | Entity ID | Type    | Department |
|---------------|-----------|---------|------------|
| Neha Mehta    | E100000   | student | CIVIL      |
| Neha Kumar    | E100001   | student | Physics    |
| Neha Singh    | E100002   | student | Admin      |
| Neha Rao      | E100005   | student | BIO        |
| Neha Malhotra | E100010   | student | Maths      |

Campus Security

Welcome back! Please sign in to continue.

Username

Enter your username

Password

\*\*\*\*\*

Log In

Login using *Password*

# "Search by Face" Identification

Advanced security feature for person identification from images

## FACE IDENTIFICATION

- Search by
- Face Image
  - Name
  - Device Hash
  - CardID
  - EntityD

TrustUsBro

DashboardCampus ActivitySearchAlertsUploadLogout

### Search Entities

Find students, faculty, or staff by a specific criterion.

Search By

Search Value

Name

neha

Search

#### Search Results

| Name          | Entity ID | Type    | Department |
|---------------|-----------|---------|------------|
| Neha Mehta    | E100000   | student | CIVIL      |
| Neha Kumar    | E100001   | student | Physics    |
| Neha Singh    | E100002   | student | Admin      |
| Neha Rao      | E100005   | student | BIO        |
| Neha Malhotra | E100010   | student | Maths      |

# Heat Maps for better understanding of Crowding

Time line Selector

Animated Heatmap

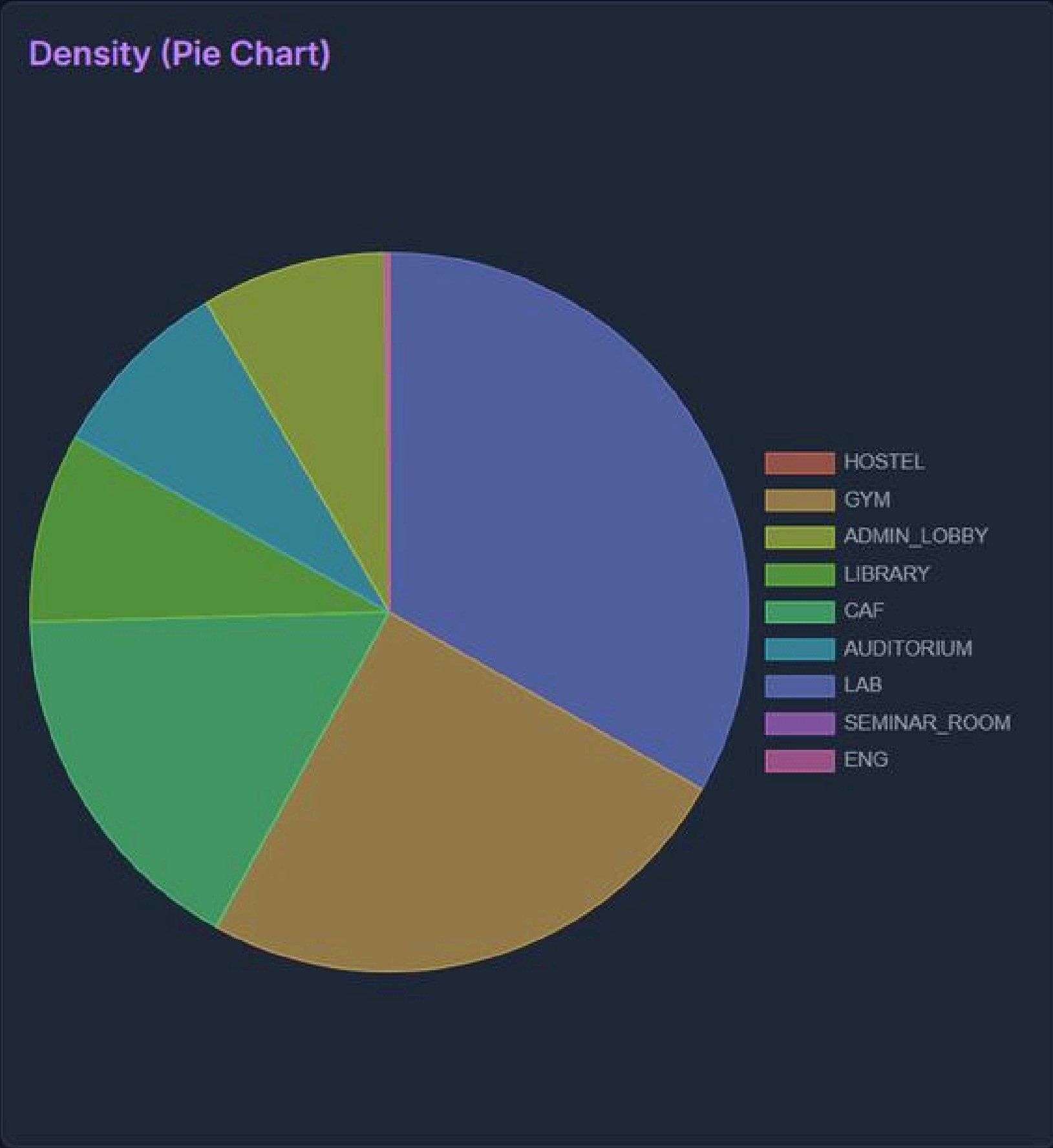
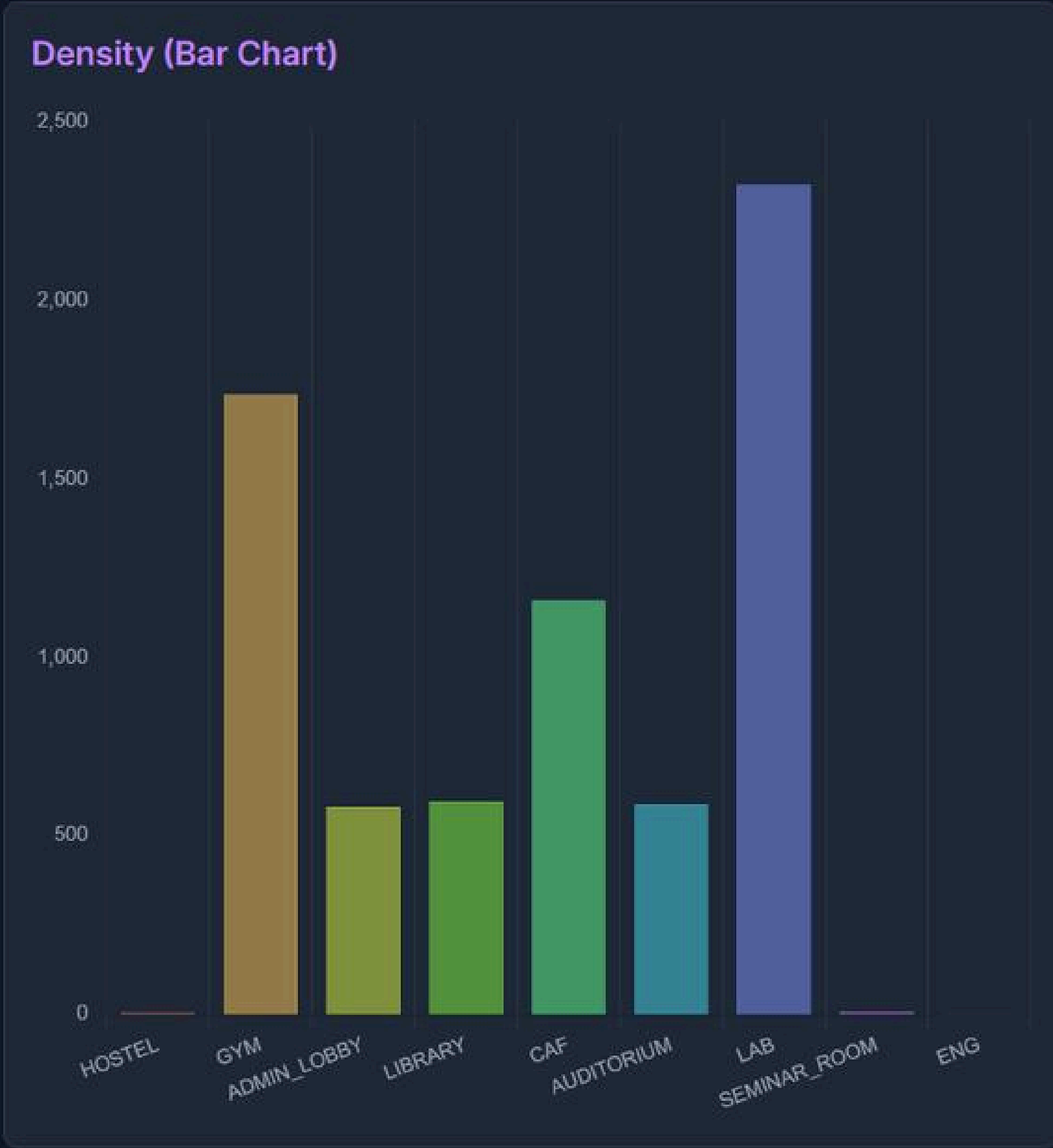
Compare Option



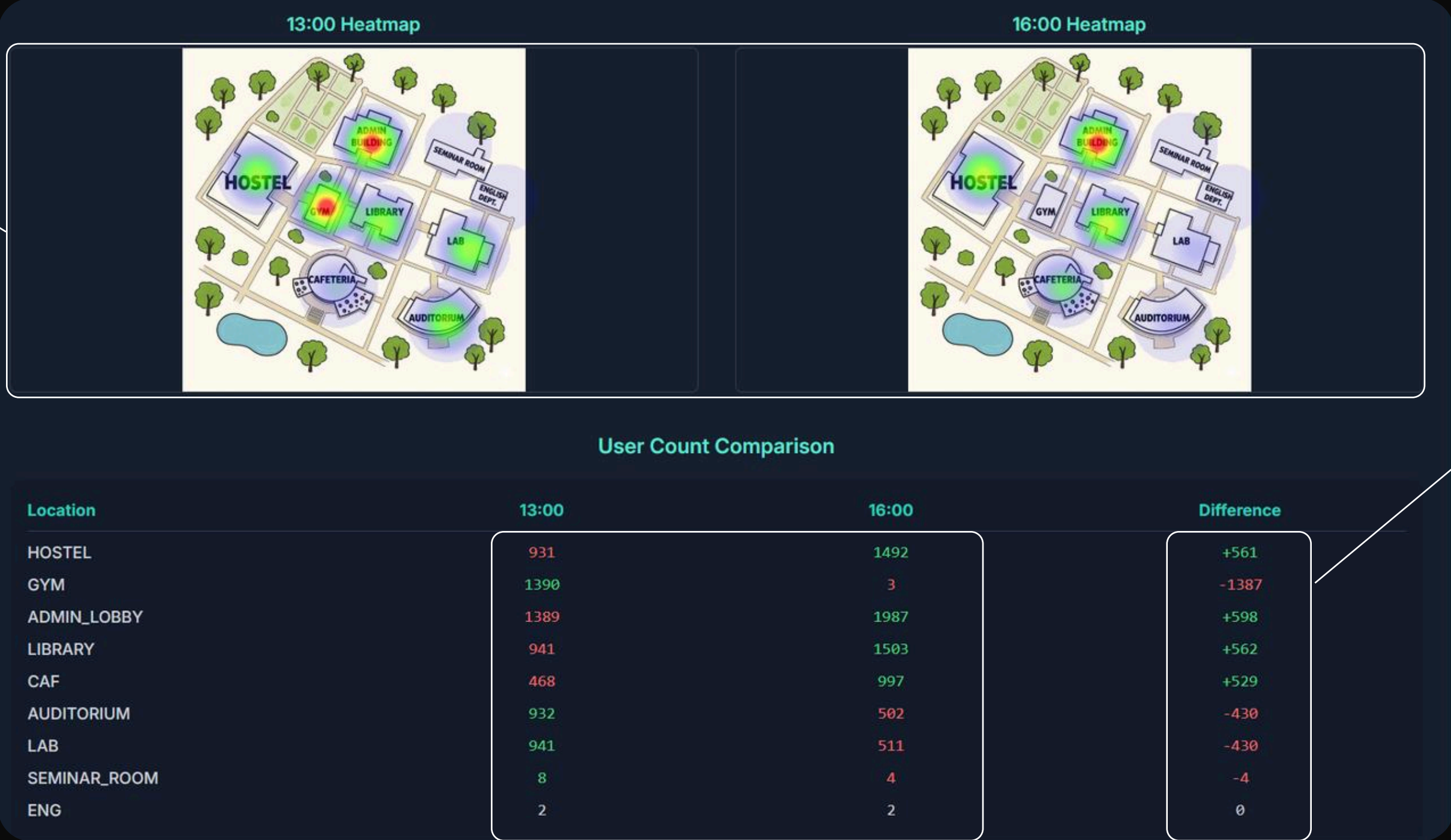
Compare Hours



# Visualization of the Heat Map data for easy interpretation



Different Time Compare



Difference

Crowd Count

# AI Integration using Gemini API for Summarizing results

### Behavioral Analysis

#### Daily Schedule Overview for Neha Gupta

##### Morning Routine (12 AM - 7 AM)

The user demonstrates an unconventional morning schedule, often active on campus during hours when most are asleep. Their activity is centered around the Auditorium and Cafeteria in the very early hours, suggesting involvement in late-night events, study groups, or performance rehearsals. This period of high activity is followed by a brief and less predictable return to the Hostel before their day formally begins with a consistent early start back at the Auditorium.

##### Daytime Activities (8 AM - 3 PM)

The main part of the day is highly structured and appears academically rigorous. The user follows a distinct, repeating pattern of moving between the Lab for focused work, the Hostel for breaks or personal time, and the Admin Lobby for what are likely meetings or administrative tasks. This predictable cycle indicates a disciplined approach to managing their core academic and university-related responsibilities.

##### Evening Schedule (4 PM - 12 AM)

The evening begins with a transition into social or personal wellness activities, typically starting at the Cafeteria and sometimes including a visit to the Gym. However, this downtime is followed by a clear shift back to academics, with time dedicated to the Library and a significant, focused work session in the Lab. The day consistently concludes with a return to the Hostel, often after a final stop at the Admin Lobby, before winding down for the night.

##### Activity Patterns

- **Most Frequent Location:** Hostel

### Campus Activity Hotspots

View predicted user density across campus for any day and time.

25-08-2025 

All Users 

Get Location Density

#### Overview of 2025-08-25 for all Users

##### High Traffic Areas

- **LAB:** Experiences its peak activity very late at night, suggesting extended study or research sessions.
- **HOSTEL:** Becomes busiest in the evening as students return for the day.
- **LIBRARY:** Sees a surge of activity later in the evening, peaking as a primary late study location.

##### Low Activity Areas

- **SEMINAR\_ROOM:** Shows very limited and sporadic use throughout the day.
- **ENG:** Remains exceptionally quiet with minimal foot traffic.

##### Movement Patterns Throughout the Day

Activity begins with a morning peak at the **GYM**, shifting to administrative areas like the **ADMIN\_LOBBY** in the afternoon. As evening approaches, the **HOSTEL** becomes the hub of activity, with traffic then moving to the **LIBRARY** and **LAB** for late-night study and work, which remain busy into the early morning hours.

**Key Insight:** Significant academic activity, particularly in the labs, extends unusually late into the night and early morning hours.

## Real Time Kafka Pipeline

A robust, event-driven architecture for real-time data ingestion.



### Pushing Data Streams

API endpoints act as Kafka Producers, pushing high-volume data streams (swipes, logs) **to dedicated topics**



### Data Never Lost

This ensures data is never lost, even if downstream services are busy or temporarily offline.



### Scalable Consumers

A dedicated consumer service subscribes to Kafka topics, processes the raw event data, and stores it efficiently in MongoDB for analysis.

# Alert Systems for getting Instant Updates about issues in the campus

Overcrowding

Violation

Missing person

TrustUsBro

DashboardCampus ActivitySearchAlertsUploadLogout

System Alerts

Active Missing4385

High Priority4385

Overcrowding3

Violations717

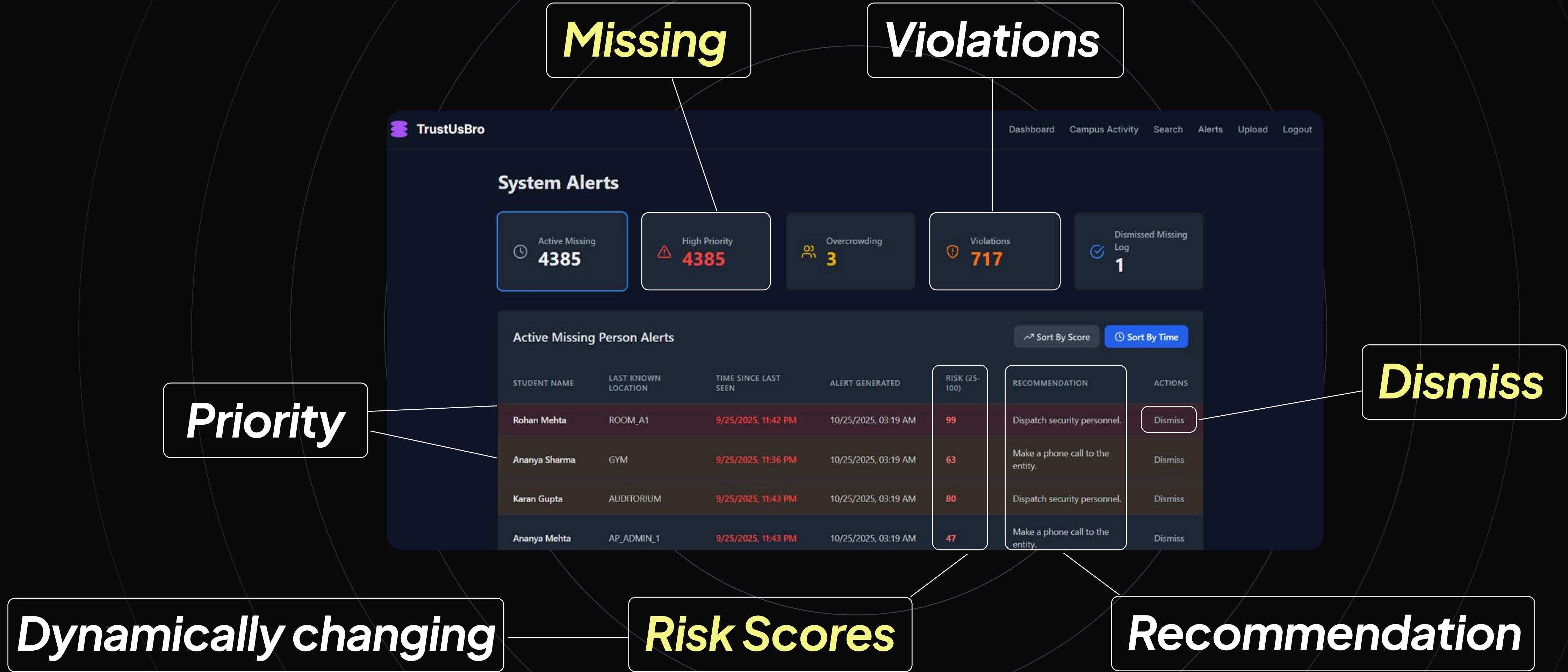
Dismissed Missing Log1

Active Missing Person Alerts

Sort By Score

Sort By Time

| STUDENT NAME  | LAST KNOWN LOCATION | TIME SINCE LAST SEEN | ALERT GENERATED      | RISK (25-100) | RECOMMENDATION                   | ACTIONS |
|---------------|---------------------|----------------------|----------------------|---------------|----------------------------------|---------|
| Rohan Mehta   | ROOM_A1             | 9/25/2025, 11:42 PM  | 10/25/2025, 03:19 AM | 99            | Dispatch security personnel.     | Dismiss |
| Ananya Sharma | GYM                 | 9/25/2025, 11:36 PM  | 10/25/2025, 03:19 AM | 63            | Make a phone call to the entity. | Dismiss |
| Karan Gupta   | AUDITORIUM          | 9/25/2025, 11:43 PM  | 10/25/2025, 03:19 AM | 80            | Dispatch security personnel.     | Dismiss |
| Ananya Mehta  | AP_ADMIN_1          | 9/25/2025, 11:43 PM  | 10/25/2025, 03:19 AM | 47            | Make a phone call to the entity. | Dismiss |





Thank  you